

Multiple Choice (10 marks)

1. Ultraviolet light of 300 nm wavelength strikes the surface of a material which has a work function of 2.0 eV. What is the velocity of the electrons emitted from the surface?
 - (a) 0.87×10^6 m/sec
 - (b) 0.72×10^6 m/sec
 - (c) 0.55×10^6 m/sec
 - (d) 0.40×10^6 m/sec
 - (e) 1.2×10^6 m/sec
2. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
 - (a) Dye laser
 - (b) Semiconductor laser
 - (c) Gas laser
 - (d) Chemical laser
 - (e) Fiber laser
3. In either a fission event or a fusion event, a quantity that remains unchanged is the
 - (a) kinetic energy
 - (b) number of electrons
 - (c) potential energy
 - (d) temperature of the nucleus
 - (e) mass
4. An ideal battery of constant voltage V supplies current to a circuit that has total resistance R . If R is doubled, what will happen to the total power dissipated by the circuit?
 - (a) It will quadruple.
 - (b) It will increase by a factor of 2.
 - (c) It will become zero.
 - (d) It will be halved.
 - (e) It will double.
5. What is the resistance of a piece of nichrome wire 225 centimeters long with a cross-sectional area of 0.015 square centimeter?

- (a) 15 ohms
- (b) 1.5 ohms
- (c) 2.25 ohms
- (d) 50 ohms
- (e) 0.15 ohms

True / False (10 marks)

Answer True or False

- 6. True or False: An increase in speed that doubles the kinetic energy of a moving body results in its momentum tripling.
- 7. True or False: The discrete orbits of electrons are best described by neutrons, which do not play a role in electron behavior within atoms.
- 8. True or False: Removing two protons and two neutrons from neon-20 results in a magnesium-18 nucleus.
- 9. True or False: Big brother and little sister can balance on a seesaw solely because their velocities are equal.
- 10. True or False: The least radiation frequency capable of positron-electron pair production is 2.47×10^{20} Hz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the phenomenon of color mixing in light, specifically analyzing how the overlap of red and blue light results in the perception of magenta, including the underlying principles of additive color theory.
- 12. Discuss how to calculate the horsepower developed by a jet aircraft engine producing a thrust of 3000 lb at a velocity of 600 mi/hr, and analyze the implications of this power output.

End of Paper